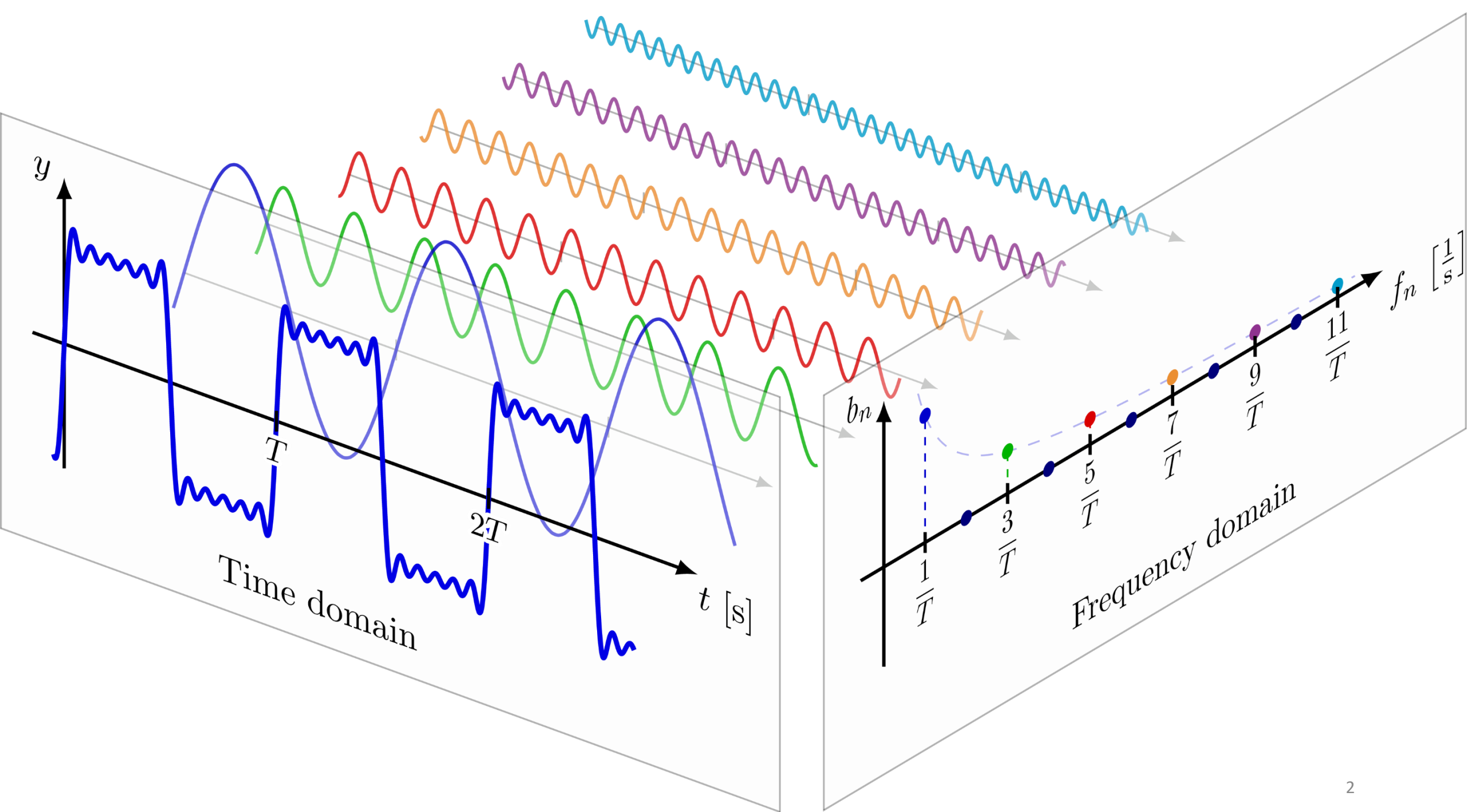


Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-9: Image Enhancement in the Frequency Domain + Restoration

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-4)

Md. Ariful Islam
Assistant Professor
Dept. of Robotics and Mechatronics Engineering
University of Dhaka



8.11 Convolution

Convolution is a mathematical operation that combines two functions to produce a third function. It is widely used in fields like signal processing, image processing, and deep learning.

The operation essentially measures how much one function overlaps with another as one is shifted over the other.

The convolution of two continuous functions $f(t)$ and $g(t)$ is defined as:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) \cdot g(t - \tau) d\tau$$

$f(t)$: This is the input function (e.g., a signal or image). It represents the data we want to process.

$g(t)$: This is the kernel or filter function. It defines the operation we want to perform on the input (e.g., blurring, edge detection).

τ : This is the dummy variable of integration. It represents the shift applied to the kernel $g(t)$ as it slides over $f(t)$.

$t-\tau$: This is the shifted version of the kernel. The kernel $g(t)$ is flipped and shifted by τ before being multiplied by $f(\tau)$.

$f(\tau) \cdot g(t-\tau)$: This is the element-wise product of the input function $f(\tau)$ and the shifted kernel $g(t-\tau)$. It measures how much the input and kernel overlap at each point.

The convolution of two discrete functions $f[n]$ and $g[n]$ is defined as:

$$(f * g)[n] = \sum_{m=-\infty}^{\infty} f[m] \cdot g[n - m]$$

8.11.1 2-D Convolution Theorem

$$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n)$$

for $x = 0, 1, 2, \dots, M - 1$ and $y = 0, 1, 2, \dots, N - 1$.

$$f(x, y) \star h(x, y) \Leftrightarrow F(u, v) H(u, v)$$

$$f(x, y) h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$$

8.12 Fourier Spectrum and Phase Angle

Because the 2-D DFT is complex in general, it can be expressed in polar form:

$$F(u, v) = |F(u, v)|e^{j\phi(u,v)}$$

where the magnitude

$$|F(u, v)| = \left[R^2(u, v) + I^2(u, v) \right]^{1/2}$$

is called the Fourier (or frequency) spectrum

$$\phi(u, v) = \arctan \left[\frac{I(u, v)}{R(u, v)} \right]$$

is the phase angle.

Finally, the power spectrum is defined as

$$\begin{aligned} P(u, v) &= |F(u, v)|^2 \\ &= R^2(u, v) + I^2(u, v) \end{aligned}$$

R and I are the real and imaginary parts of (u, v) and all computations are carried out for the discrete variables

$$u = 0, 1, 2, \dots, M - 1 \qquad v = 0, 1, 2, \dots, N - 1$$

Therefore,

$|F(u, v)|$, $\phi(u, v)$, and $P(u, v)$ are arrays of size $M \times N$

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-i2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

$$F(0, 0) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)$$

which indicates that the zero-frequency term is proportional to the average value of $f(x, y)$. That is,

$$\begin{aligned} F(0, 0) &= \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \\ &= \overline{f}(x, y) \end{aligned}$$

where $\overline{f}$ denotes the average value of f . Then,

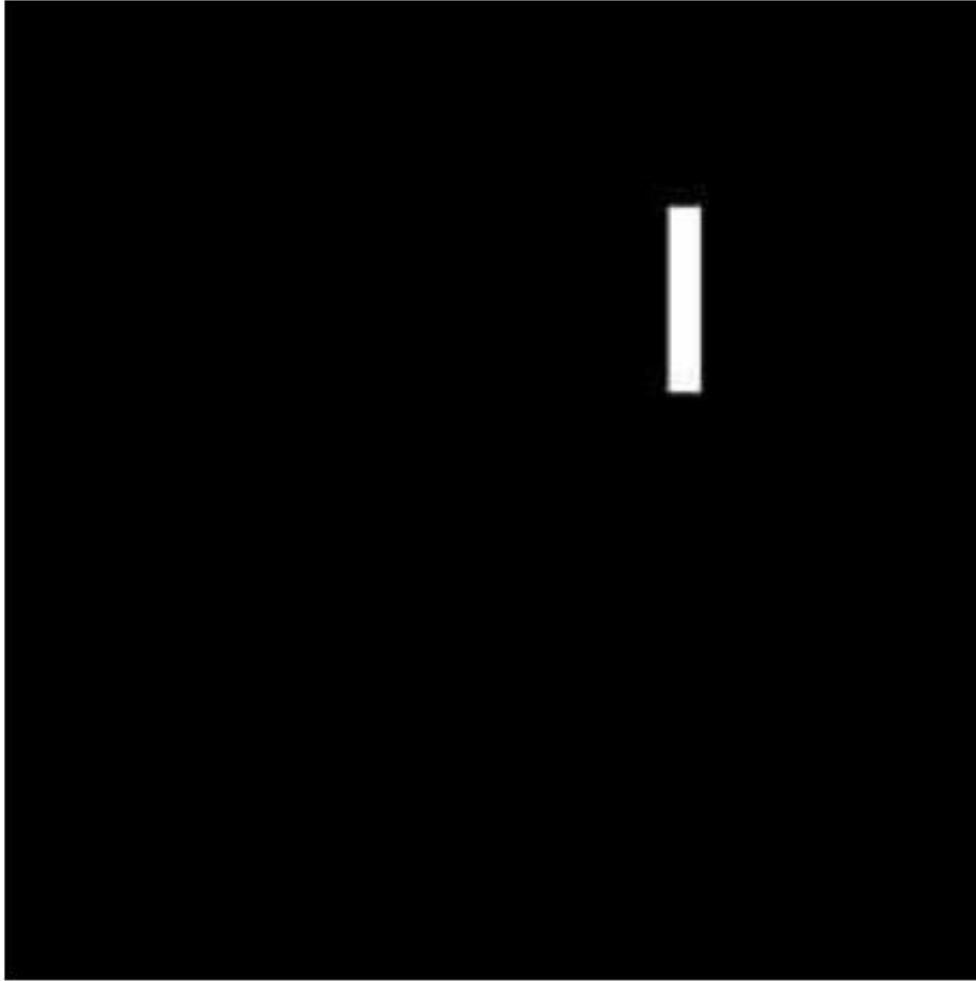
$$|F(0, 0)| = MN |\overline{f}(x, y)|$$

Because the proportionality constant MN usually is large, typically is the largest component of the spectrum by a factor that can be several orders of magnitude larger than other terms.

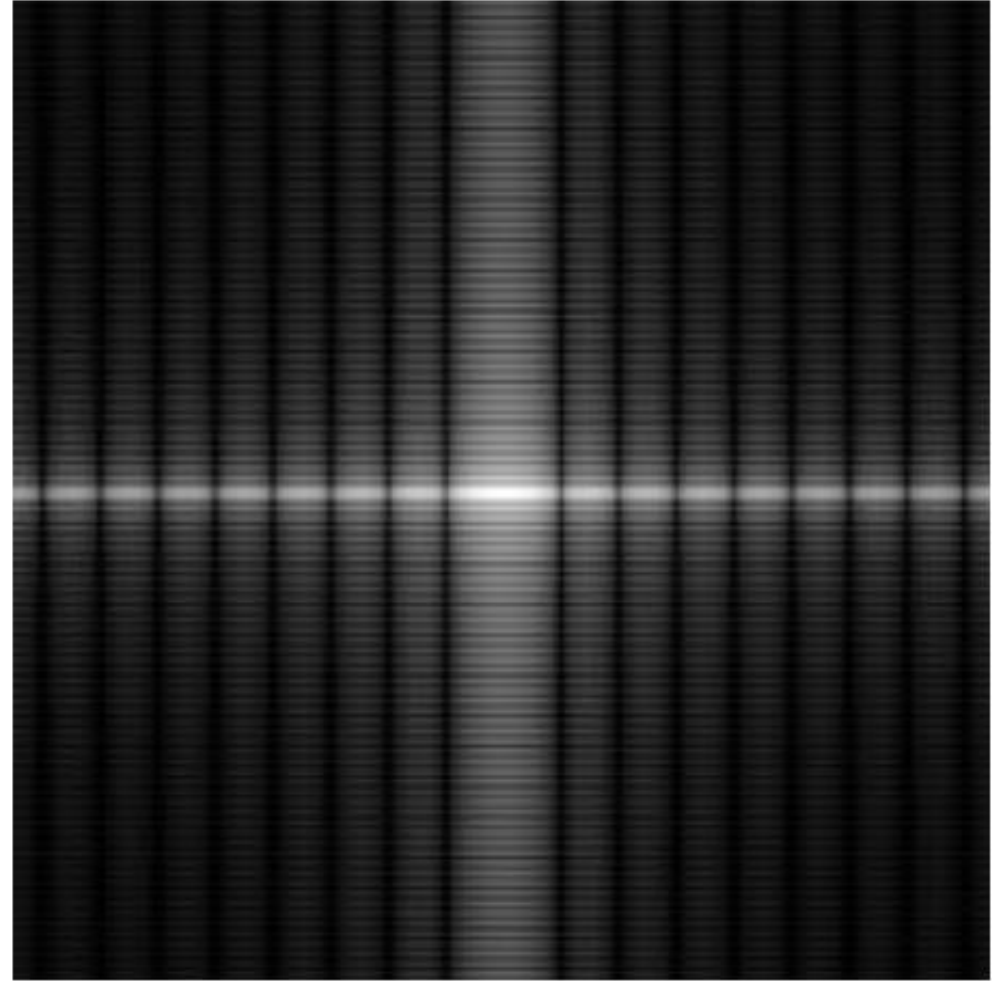
Because frequency components u and v are zero at the origin, $(0, 0)$ sometimes is called the dc component of the transform.

This terminology is from electrical engineering, where “dc” signifies direct current (i.e., current of zero frequency).

Example: Figure (a) shows a simple image and Fig. (b) shows its spectrum, whose values were scaled to the range $[0, 255]$ and displayed in image form.



(a)

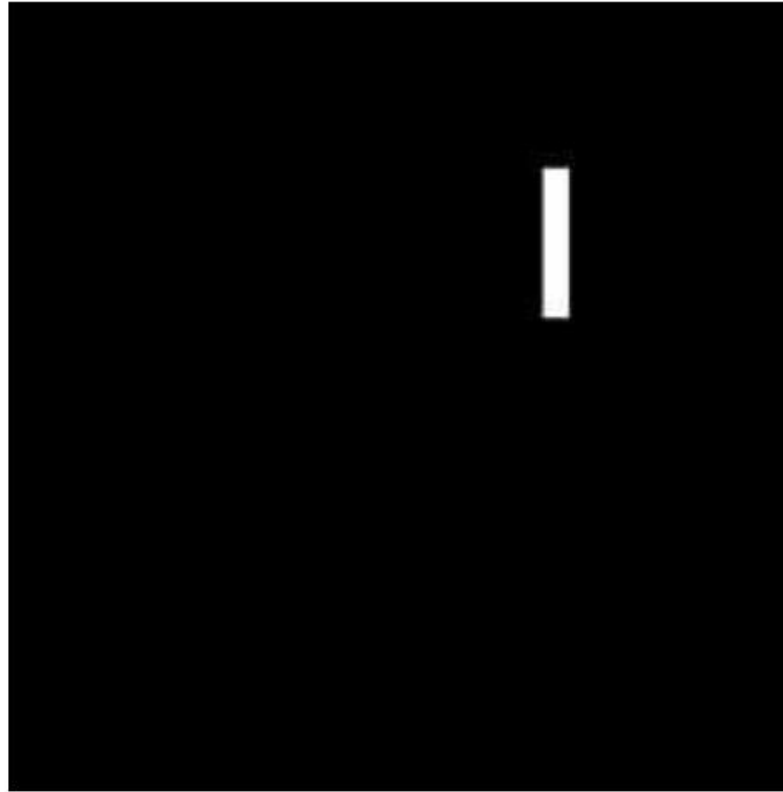


(b)

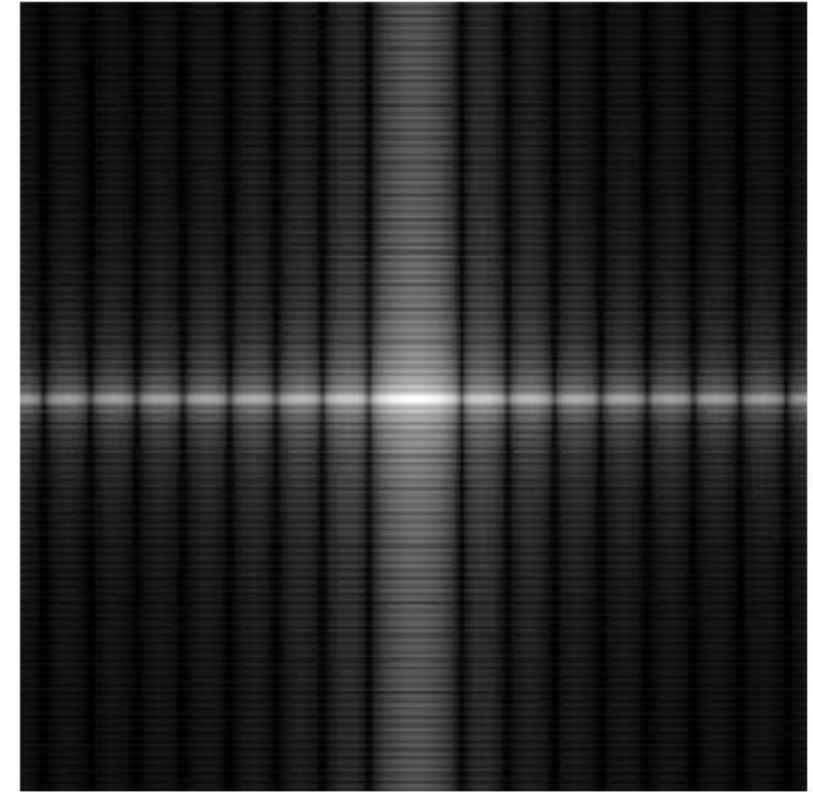
The origins of both the spatial and frequency domains are at the top left.

Two things are apparent in Fig. (b).

As expected, the area around the origin of the transform contains the highest values (and thus appears brighter in the image).

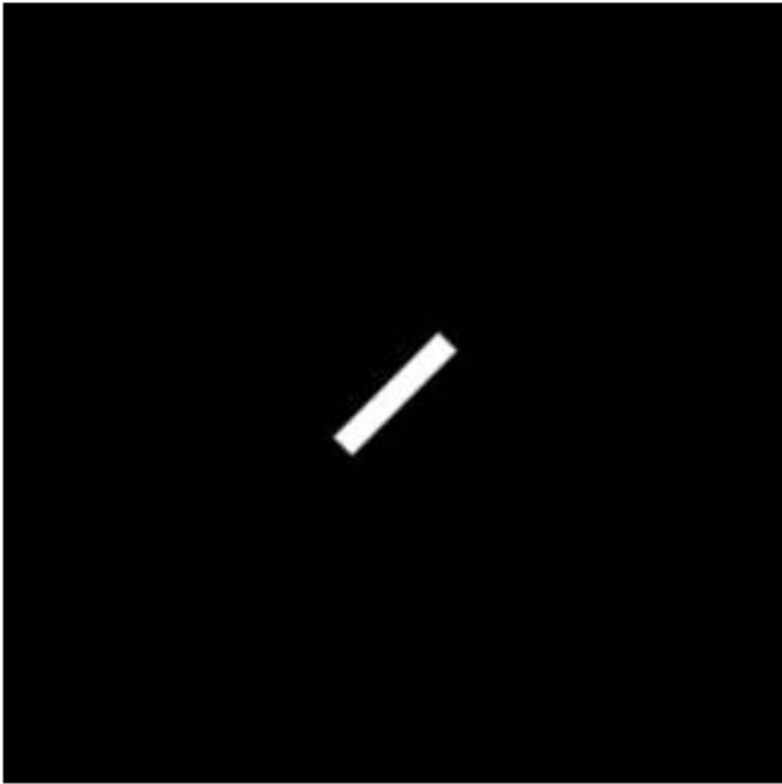


(a)

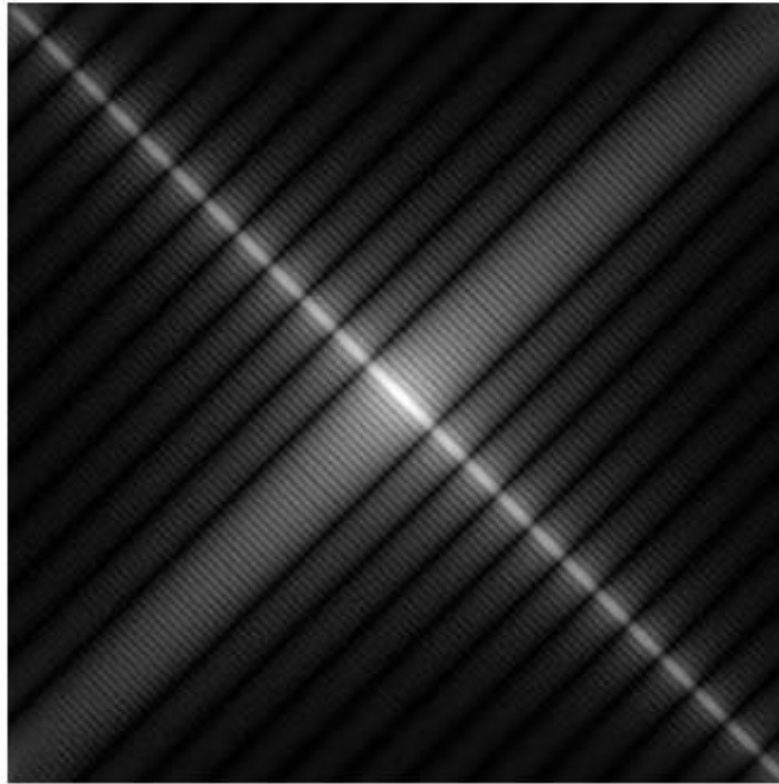


(b)

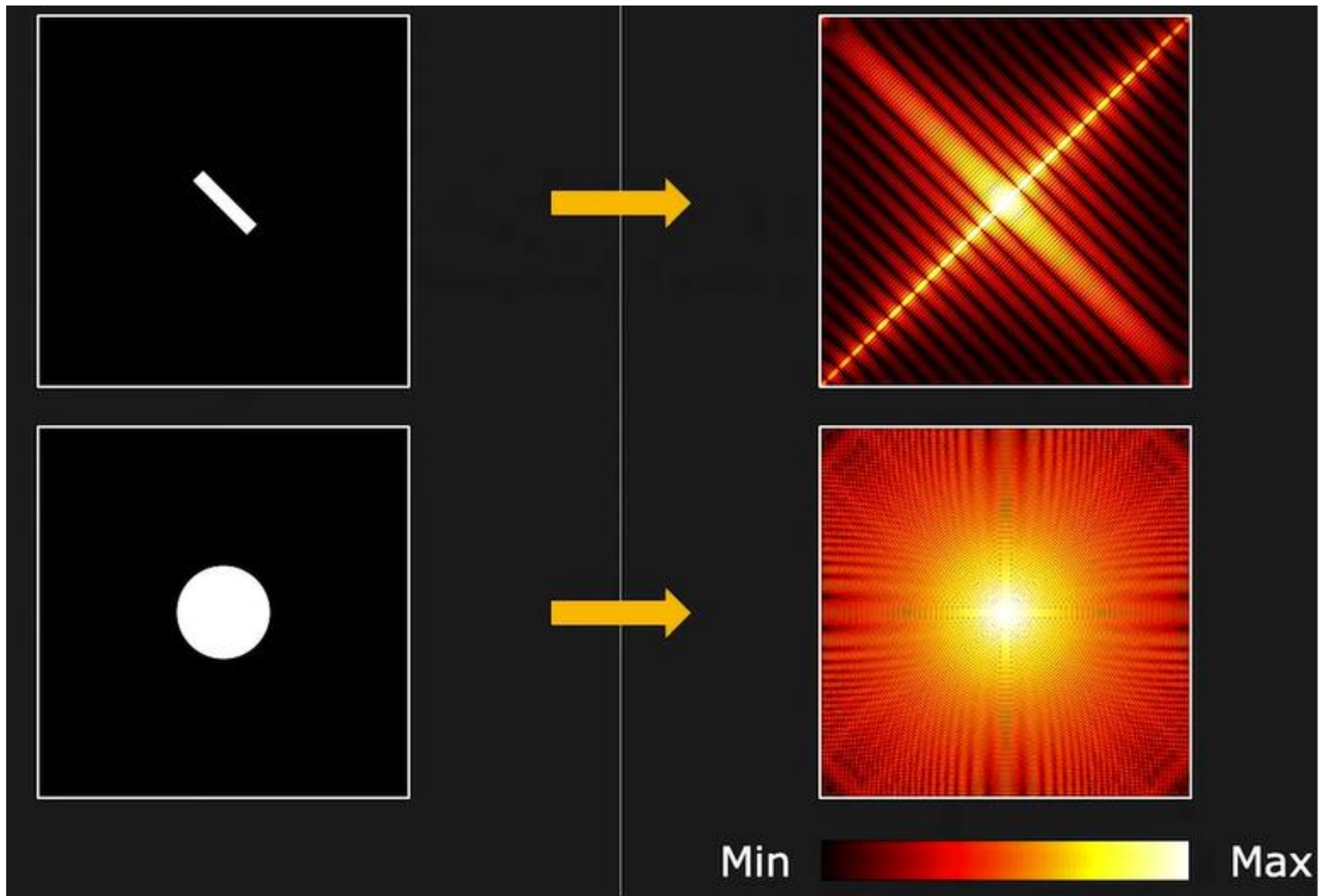
However, note that the four corners of the spectrum contain similarly high values. The reason is the periodicity property



Rotated rectangle



The corresponding spectrum. The spectrum corresponding to the translated rectangle is identical to the spectrum corresponding to the original image



9. Image Restoration

Image restoration focuses on removing noise, blur, or other artifacts from an image to recover the original, clean image.

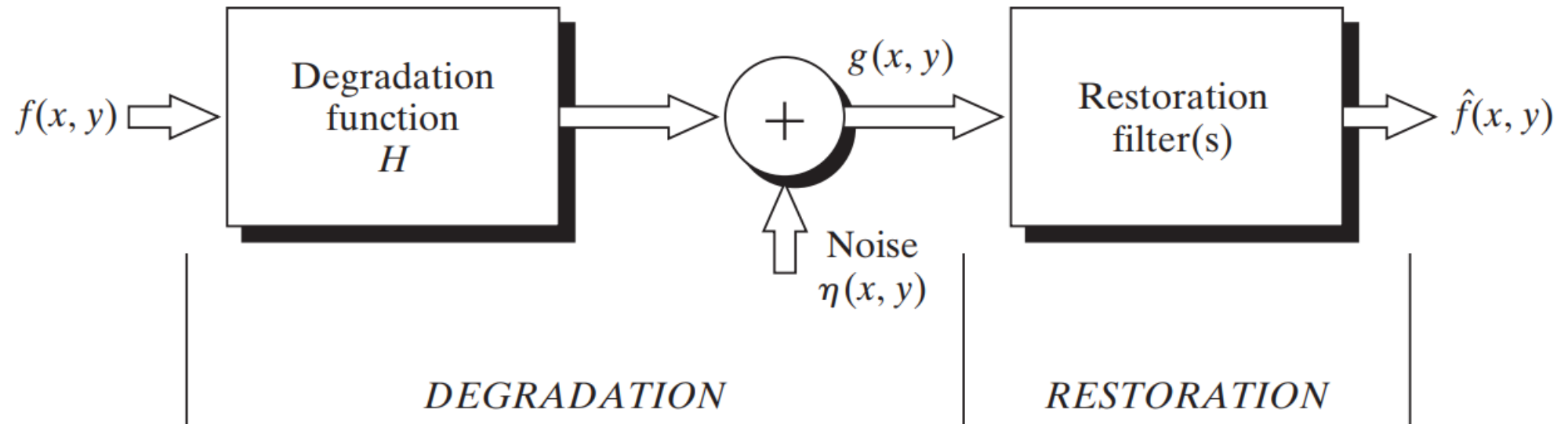
Common causes of degradation include:

Noise: Random variations in brightness or color (e.g., Gaussian noise, salt-and-pepper noise).

Blur: Caused by camera shake, motion, or out-of-focus lenses.

Compression artifacts: Lossy compression (e.g., JPEG) can introduce blocky or pixelated artifacts.

Given $g(x,y)$, some knowledge about the degradation function H , and some knowledge about the additive noise term $\eta(x, y)$ the objective of restoration is to obtain an estimate $\hat{f}(x, y)$ of the original image.



if H is a linear, position-invariant process, then the degraded image is given in the spatial domain by

$$g(x, y) = h(x, y) \star f(x, y) + \eta(x, y)$$

Where $h(x,y)$ is the spatial representation of the degradation function and, as in, the symbol $*$ indicates convolution.

The convolution in the spatial domain is analogous to multiplication in the frequency domain

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

9.1 Noise Models

The principal sources of noise in digital images *arise during image acquisition and/or transmission.*

The performance of imaging *sensors is affected by a variety of factors,* such as

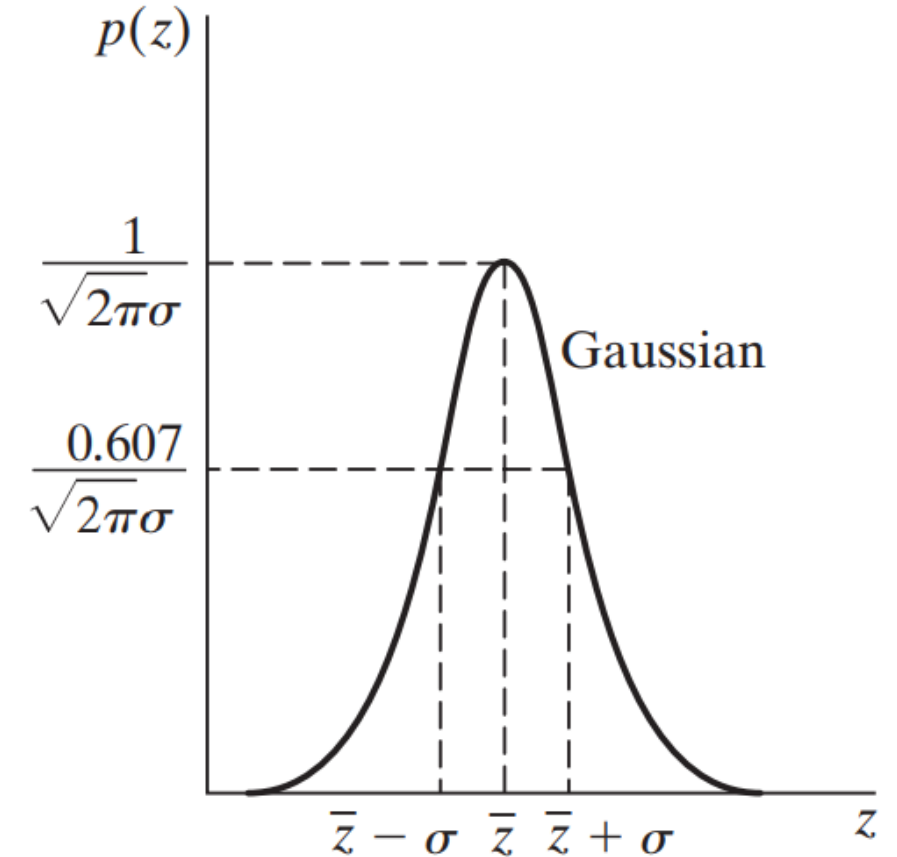
- ❑ *environmental conditions* during image acquisition, and
- ❑ by the *quality of the sensing elements* themselves.

Some important Noise Probability Density Functions

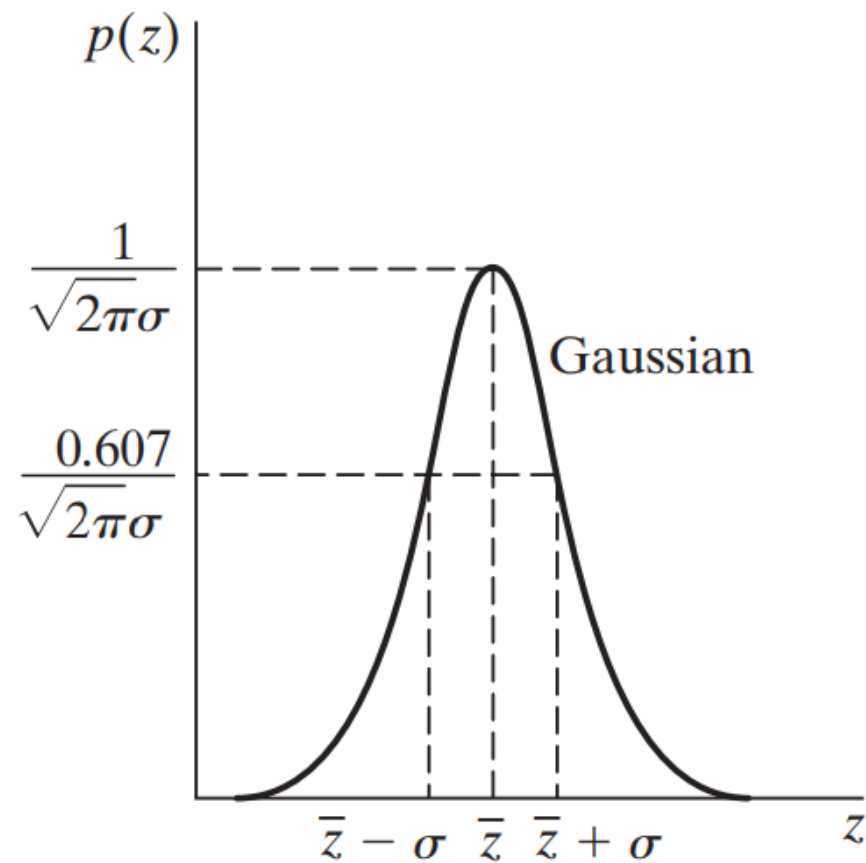
9.1.1 Gaussian noise

The PDF of a Gaussian random variable, z , is given by

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-\bar{z})^2/2\sigma^2}$$



where z represents intensity, $\bar{z}$ is the mean[†] (average) value of z , and σ is its standard deviation. The standard deviation squared, σ^2 , is called the *variance* of z .



approximately 70% of its values will be in the range $[(\bar{z} - \sigma), (\bar{z} + \sigma)]$, and about 95% will be in the range $[(\bar{z} - 2\sigma), (\bar{z} + 2\sigma)]$.

9.1.2 Rayleigh noise

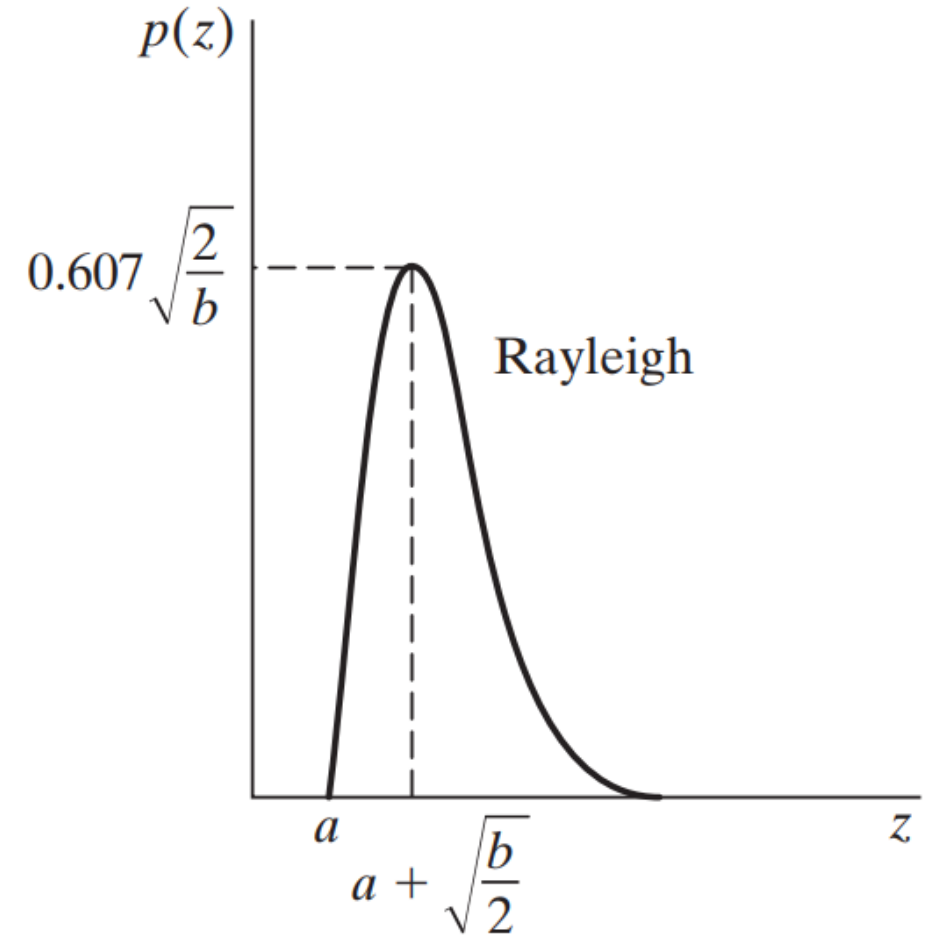
The PDF of Rayleigh noise is given by

$$p(z) = \begin{cases} \frac{2}{b}(z - a)e^{-(z-a)^2/b} & \text{for } z \geq a \\ 0 & \text{for } z < a \end{cases}$$

The mean and variance of this density are given by

$$\bar{z} = a + \sqrt{\pi b/4}$$

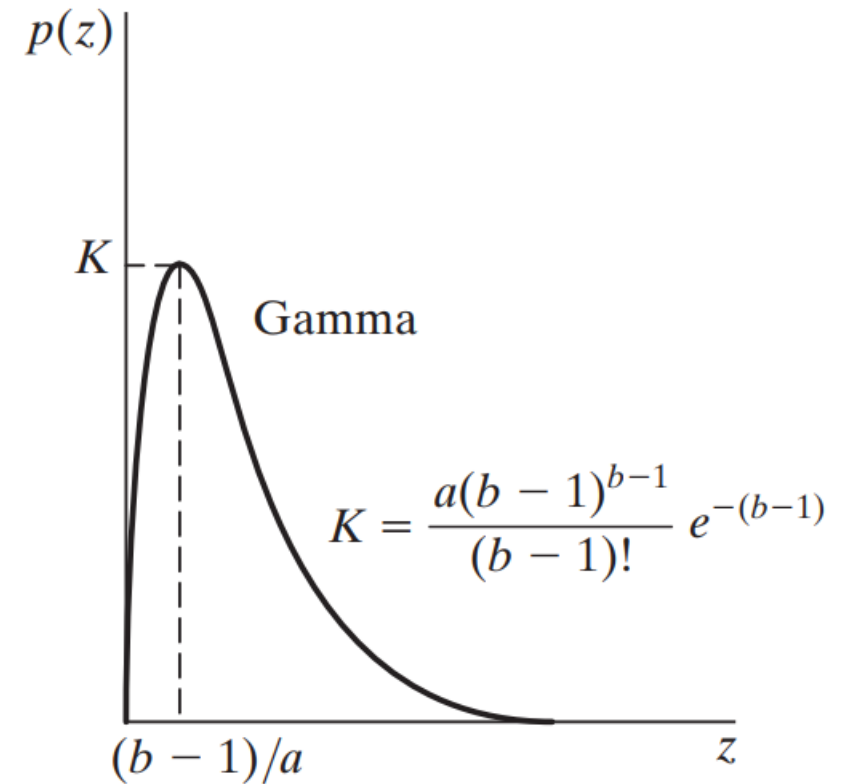
$$\sigma^2 = \frac{b(4 - \pi)}{4}$$



9.1.3 Erlang (gamma) noise

The PDF of Erlang noise is given by

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$



where the parameters are such that $a > 0$, b is a positive integer, and “!” indicates factorial. The mean and variance of this density are given by

$$\bar{z} = \frac{b}{a} \quad \sigma^2 = \frac{b}{a^2}$$

9.1.4 Exponential noise

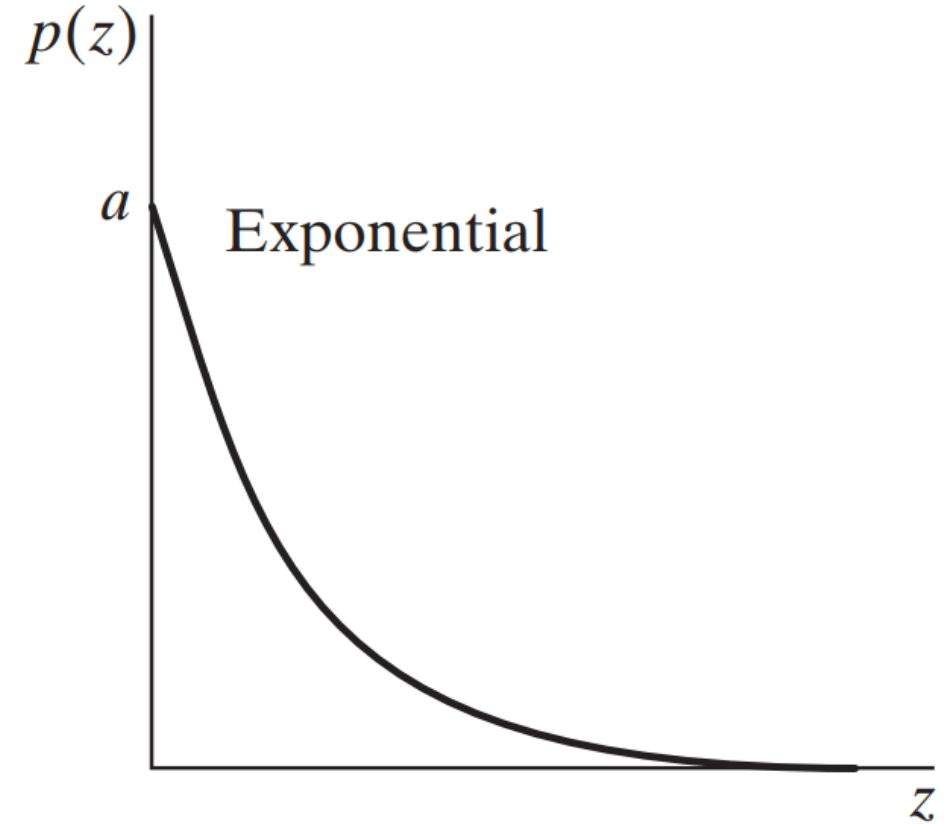
The PDF of exponential noise is given by

$$p(z) = \begin{cases} ae^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

Where $a > 0$.

The mean and variance of this density function are

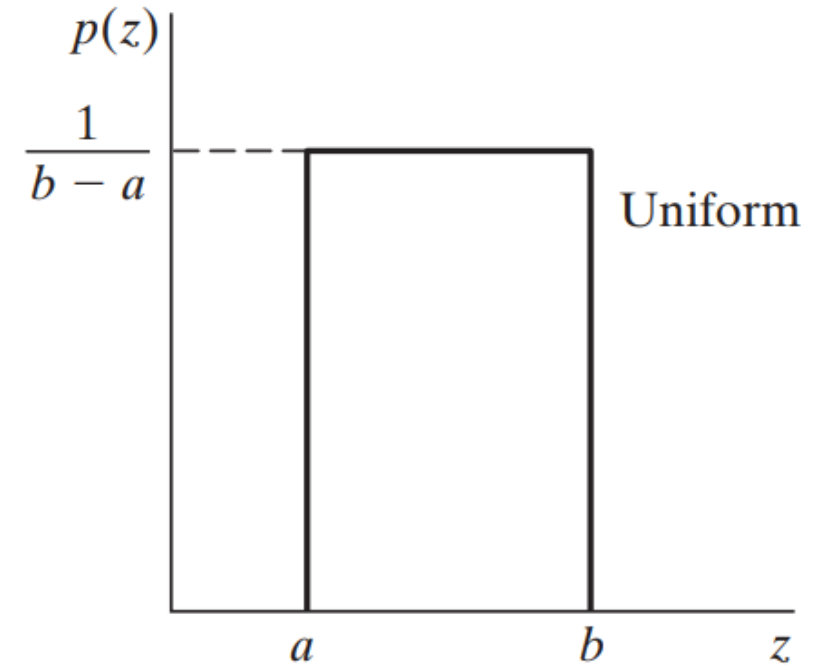
$$\bar{z} = \frac{1}{a} \quad \sigma^2 = \frac{1}{a^2}$$



9.1.5 Uniform noise

The PDF of uniform noise is given by

$$p(z) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq z \leq b \\ 0 & \text{otherwise} \end{cases}$$



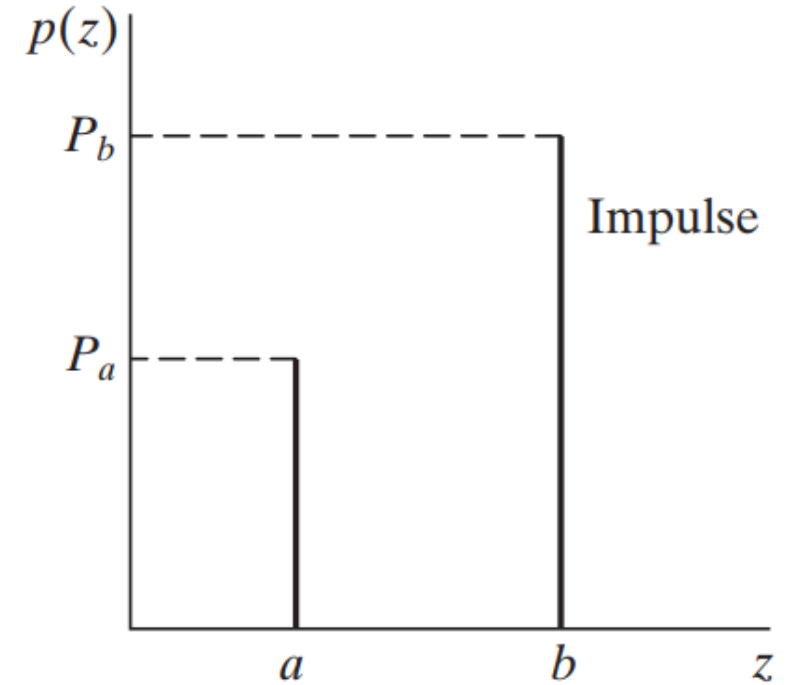
The mean of this density function is given by $\bar{z} = \frac{a+b}{2}$

and its variance by $\sigma^2 = \frac{(b-a)^2}{12}$

9.1.6 Impulse (salt-and-pepper) noise

The PDF of (bipolar) impulse noise is given by

$$p(z) = \begin{cases} P_a & \text{for } z = a \\ P_b & \text{for } z = b \\ 0 & \text{otherwise} \end{cases}$$

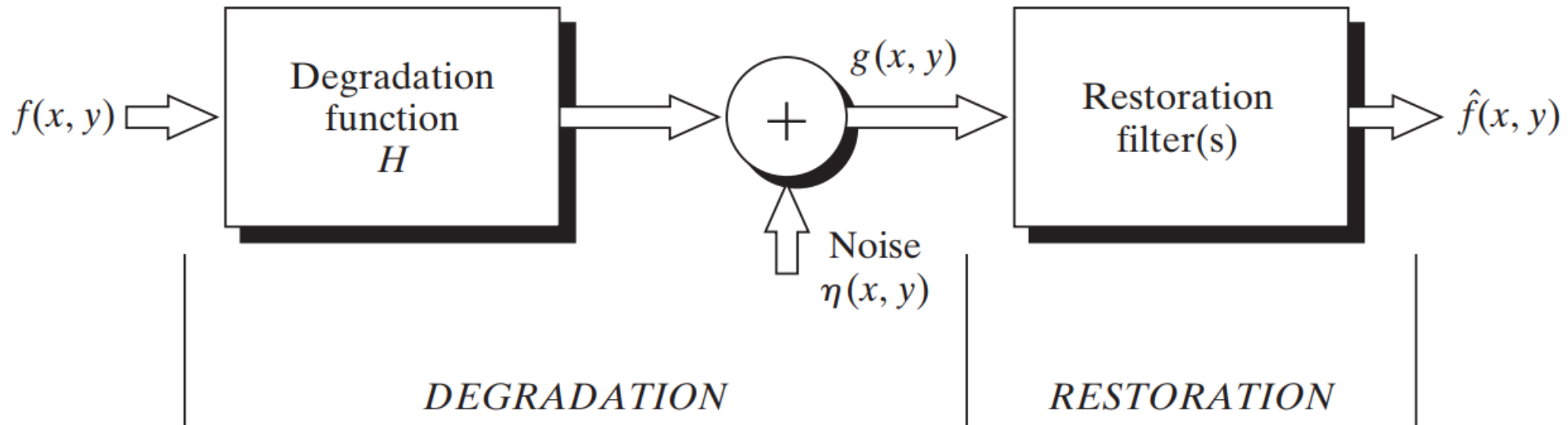


If $b > a$, intensity b will appear as a light dot in the image. Conversely, level a will appear like a dark dot. If either P_a or P_b is zero, the impulse noise is called unipolar.

9.2 Linear, Position-Invariant Degradations

The input-output relationship before the restoration stage is expressed as

$$g(x, y) = H[f(x, y)] + \eta(x, y)$$



For the moment, let us assume that

$$\eta(x, y) = 0 \text{ so that } g(x, y) = H[f(x, y)].$$

H is linear if

$$H[af_1(x, y) + bf_2(x, y)] = aH[f_1(x, y)] + bH[f_2(x, y)]$$

where a and b are scalars and $f_1(x, y)$ and $f_2(x, y)$ are any two input images.

If $a=b=1$

$$H[f_1(x, y) + f_2(x, y)] = H[f_1(x, y)] + H[f_2(x, y)]$$

which is called the property of additivity.

This property simply says that, if H is a linear operator, the response to a sum of two inputs is equal to the sum of the two responses.

With $f_2(x,y) = 0$

$$H[af_1(x, y)] = aH[f_1(x, y)]$$

which is called the property of homogeneity.

- It says that the response to a constant multiple of any input is equal to the response to that input multiplied by the same constant.

Thus a linear operator possesses both the property of additivity and the property of homogeneity.

An operator having the input-output relationship $g(x, y) = H[f(x, y)]$ is said to be position (or space) invariant if

$$H[f(x - \alpha, y - \beta)] = g(x - \alpha, y - \beta)$$

for any $f(x, y)$ and any α and β .

This definition indicates that the response at any point in the image depends only on the value of the input at that point, not on its position.

9.3 Degradation model in case of continuous function

Delta function:

$$\delta(x, y) = \begin{cases} 1 & x = 0 \text{ \& } y = 0 \\ 0 & \textit{otherwise.} \end{cases}$$

Shifted Delta function:

$$\delta(x - x_0, y - y_0) = \begin{cases} 1 & x = x_0 \text{ \& } y = y_0 \\ 0 & \textit{otherwise.} \end{cases}$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(x, y) dx dy = 1$$

If we multiply an image $f(x, y)$ with the delta function

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \cdot \delta(x, y) dx dy = f(0, 0)$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0)$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \cdot \delta(x - x_0, y - y_0) dx dy = f(x_0, y_0) \dots (1)$$

With a slight change in notation in the definition of the impulse in equation (1), $f(x, y)$ can be expressed as

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) \delta(x - \alpha, y - \beta) d\alpha d\beta \dots (2)$$

$$g(x, y) = H[f(x, y)] + \eta(x, y) \quad \text{if } \eta(x, y) = 0$$

$$g(x, y) = H \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) \delta(x - \alpha, y - \beta) d\alpha d\beta \right]$$

If H is a linear operator

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} H[f(\alpha, \beta) \delta(x - \alpha, y - \beta)] d\alpha d\beta \dots\dots(3)$$

as, $f(\alpha, \beta)$ is independent of x, and y. Therefore the equation (3) can be rewritten as

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) H[\delta(x - \alpha, y - \beta)] d\alpha d\beta$$

The term,

$$h(x, \alpha, y, \beta) = H[\delta(x - \alpha, y - \beta)]$$

is called the impulse response of H.

if $\eta(x, y) = 0$ in the following equation.

$$g(x, y) = H[f(x, y)] + \eta(x, y)$$

then, $h(x, \alpha, y, \beta)$ is the response of H to an impulse at coordinates (x, y) .

In optics, the impulse becomes a point of light and $h(x, \alpha, y, \beta)$ is commonly referred to as the point spread function (PSF).

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) h(x, \alpha, y, \beta) d\alpha d\beta \dots (4)$$

This equation is known as the superposition (or Fredholm) integral of the first kind.

It is states that if impulse response of H is known, the response to any input $f(\alpha, \beta)$ can be calculated by the equation (4).

If H is position invariant,

$$H[f(x - \alpha, y - \beta)] = g(x - \alpha, y - \beta)$$

$$H[\delta(x - \alpha, y - \beta)] = h(x - \alpha, y - \beta)$$

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) h(x - \alpha, y - \beta) d\alpha d\beta \dots (5)$$

Considering the noise term, $\eta(x, y)$

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) h(x - \alpha, y - \beta) d\alpha d\beta + \eta(x, y)$$

This is the general expression of degradation model. Here we assume the degradation function, H is linear and position invariant. Many of the degradation operations in real applications can be approximated by linear position invariant model.

Using the familiar notation for convolution, we can write

$$g(x, y) = h(x, y) \star f(x, y) + \eta(x, y)$$

based on the convolution theorem, we can express it in the frequency domain as

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

9.3.1 Discrete formulation of this degradation model

1D Case: *Initially, $\eta(x, y) = 0$*

$f(x) \rightarrow$ *input function* $\rightarrow$ *Array of dimension A*

$h(x) \rightarrow$ *Degradation function* $\rightarrow$ *Array of dimension B*

$\Downarrow$

Uniformly sampled these two functions (for discretization)

$$\left. \begin{array}{l} f(x) : x = 0, 1, \dots, A-1 \\ h(x) : x = 0, 1, \dots, B-1 \end{array} \right\} \Rightarrow M$$

✓ Both of them to be made same dimension, M (adding additional number of zeros)

✓ After adding, we assume both $f(x)$ and $h(x)$ are periodic with period, M.

New array (extended version), $f_e(x)$, and $h_e(x)$

$$f_e(x) = \begin{cases} f(x) & \text{for } 0 \leq x < A \\ 0 & \text{for } A \leq x \leq M-1 \end{cases}$$

$$h_e(x) = \begin{cases} h(x) & \text{for } 0 \leq x < B \\ 0 & \text{for } B \leq x \leq M-1 \end{cases}$$

$$g_e(x) = \sum_{m=0}^{M-1} f_e(x)h_e(x-m); \quad x = 0,1,\dots,M-1$$

$\Downarrow$
Matrix form

$$g = Hf$$

Here, g and f are M-dimensional vectors.

$$f = \begin{bmatrix} f_e(0) \\ f_e(1) \\ . \\ . \\ f_e(M-1) \end{bmatrix}, \quad g = \begin{bmatrix} g_e(0) \\ g_e(1) \\ . \\ . \\ g_e(M-1) \end{bmatrix}, \text{ and } H = M \times M (\text{dimension})$$

$$H = \begin{bmatrix} h_e(0) & h_e(-1) & h_e(-2) \dots h_e[-(M-1)] \\ h_e(1) & h_e(0) & h_e(-1) \dots h_e[-(M-2)] \\ h_e(2) & h_e(1) & h_e(0) \dots h_e[-(M-3)] \\ . & & \\ . & & \\ h_e(M-1) & h_e(M-2) & h_e(M-3) \dots h_e(0) \end{bmatrix}$$

as $h_e(x)$ is periodic with period M

$$\text{so, } h_e(x + M) = h_e(x)$$

Rewrite, the matrix H :

$$H = \begin{bmatrix} h_e(0) & h_e(M-1) & h_e(M-2) \dots h_e(1) \\ h_e(1) & h_e(0) & h_e(M-1) \dots h_e(2) \\ h_e(2) & h_e(1) & h_e(0) \dots h_e(3) \\ \cdot & & \\ \cdot & & \\ h_e(M-1) & h_e(M-2) & h_e(M-3) \dots h_e(0) \end{bmatrix}$$

2D Case:

$f(x, y) \rightarrow \text{sampled by } A \times B$
 $h(x, y) \rightarrow \text{sampled by } C \times D$ } Add number of zeros to make the both are same size.

$f_e(x, y)$
 $h_e(x, y)$ } $M \times N$ Periodic in x-direction of period, M and y-direction of period, N.

The convolution expression in 2D:

$$g_e(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f_e(m, n) h_e(x - m, y - n)$$

$$x = 0, 1, \dots, M - 1$$

$$y = 0, 1, \dots, N - 1$$

Rewrite in the form of matrix with additive noise:

$$g = Hf + \eta$$

- By concatenating different rows of 2D function $f(x,y)$.

$$f \rightarrow (\text{size}) MN$$

$$H \rightarrow (\text{size}) MN \times MN$$

$$g \rightarrow (\text{size}) MN$$

$$H = \begin{bmatrix} H_0 & H_{M-1} \dots H_1 \\ H_1 & H_0 \dots H_2 \\ \vdots & \\ \vdots & \\ H_{M-1} & H_{M-2} \dots H_0 \end{bmatrix}$$

Each element, H_j of this matrix H is a matrix of dimension $N \times N$.

where the, H_j is generated from the j^{th} row of the degradation function $h(x,y)$. matrix of dimension $N \times N$.

$\Rightarrow$ *Block circulant matrix*

$$H_j = \begin{bmatrix} h_e(j,0) & h_e(j, N-1) \dots\dots\dots h_e(j,1) \\ h_e(j,1) & h_e(j,0) \dots\dots\dots h_e(j,2) \\ \cdot & \\ \cdot & \\ h_e(j, N-1) & h_e(j, N-2) \dots\dots\dots h_e(j,0) \end{bmatrix}$$

$\Rightarrow$ *Circulant matrix*

For digital image (degradation model):

$$g = Hf + \eta$$

$$f \rightarrow (M \times N)$$

$$H \rightarrow (M \times N) \times (M \times N)$$

$$g \rightarrow (M \times N)$$